An Exact Cooperation Formula for Introspection Dynamics in the Heterogeneous Public Goods Game

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April 16, 2026

Abstract

We study introspection dynamics on multiplayer games in which the payoff difference Δf_i a player evaluates when considering a strategy switch is independent of all other players' current actions, a property we call *state-independence*. Under this condition the introspection Markov chain decomposes as a random-scan product of N independent two-state chains, one per player, and the stationary distribution is a product measure. As our main application we consider the heterogeneous public goods game, where N players may differ in their contributions α_i, public goods multipliers r_i, and choice intensities β_i. We prove that the linear payoff structure implies state-independence, and the long-run cooperation probability admits the exact closed form

$$p_C^{\text{intro}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{1-\mu_{i0}-\mu_{i1}}{1+e^{\beta_i\alpha_i(1-r_i/N)}}+\mu_{i0}\right],$$

with no asymptotic approximation. Several structural consequences follow immediately: a player-specific cooperation threshold at $r_i = N$ (under symmetric mutation), payoff-neutrality under zero choice intensity, and the sign of each player's sensitivity to their own parameters.

1 Introduction

The public goods game is a canonical model of cooperation in evolutionary game theory: players decide whether to contribute to a common resource, which is multiplied and redistributed equally, creating a tension between individual incentive and collective benefit [5]. Classical analyses assume a homogeneous population, but real groups are heterogeneous: players differ in their productivity, their contributions, and how sensitively they respond to payoff differences [4].

A natural learning rule for such settings is *introspection dynamics*, introduced in [1]: at each time step a player compares their current payoff to the payoff they *would have received* had they played the alternative action (keeping all other players fixed), and switches with a probability governed by this difference. Because the comparison is self-referential rather than imitative, introspection applies to arbitrary asymmetric games where role-model imitation is unavailable or unnatural.

In [2] the authors extended introspection dynamics to multiplayer asymmetric games and derived a formula for the stationary distribution that can be evaluated numerically for any game. In their general framework the transition rate for player i switching depends on the actions of all other players through the payoff function, so the stationary distribution must be computed by solving a linear system of dimension 2^N; no closed form is available in general.

In this paper we show that whenever the payoff difference Δf_i a player evaluates when considering a switch is independent of all other players' current actions, a property we call *state-independence*, the introspection chain decomposes exactly as a product of N independent two-state chains (one per player), the stationary distribution is a product measure, and the cooperation probability p_C^{intro} admits an explicit closed form scaling as $O(N)$ rather than $O(2^N)$. We then show that the heterogeneous public goods game (PGG) satisfies state-independence as a consequence of the *linearity* of its payoff structure, yielding the exact formula (8).

The paper is organised as follows. Section 2 introduces the general framework and the state-independence condition. Section 3 proves, for any state-independent game, that individual cooperation probabilities are given by (4) and the stationary distribution is a product measure. Section 4 establishes state-independence for the PGG and derives the exact cooperation formula. Section 5 records several structural corollaries. Section 6 concludes.

2 Setup and notation

Consider N ordered players, each choosing from two actions $\{C, D\}$ coded as $\{1, 0\}$. The state space is $S = \{0,1\}^N$ with $|S| = 2^N$. Each player i has a payoff function $f_i : S \to \mathbb{R}$ and player-specific choice intensity $\beta_i > 0$. Writing $\mathbf{a}_{i\to\bar{a}_i}$ for the state with player i's action flipped to $\bar{a}_i = 1 - a_i$, the *payoff difference* for player i at state $\mathbf{a}$ is

$$\Delta f_i(\mathbf{a}) := f_i(\mathbf{a}) - f_i(\mathbf{a}_{i\to\bar{a}_i}).$$

A payoff function f_i is *state-independent* if $\Delta f_i(\mathbf{a})$ depends only on a_i, not on $\mathbf{a}_{-i}$. When this holds there is a constant $\delta_i \in \mathbb{R}$ such that

$$\Delta f_i(\mathbf{a}) = (1 - 2a_i)\,\delta_i, \tag{1}$$

where $\delta_i = \Delta f_i\big|_{a_i=0}$ is the payoff difference when player i defects. When $a_i = 0$, (1) follows directly from the definition of δ_i; when $a_i = 1$, $\Delta f_i\big|_{a_i=1} = -\Delta f_i\big|_{a_i=0} = -\delta_i$, giving (1).

Note that not all games lead to *state-independent* payoff functions. For example in [1] two of the 2 player, 2 strategy games considered to illustrate introspection dynamics are: the Prisoner's Dilemma (2 - M_1) and the Stag-hunt (2 - M_2) for $b > c_1 \leq c_2 > 0$.

$$M_1 = \begin{pmatrix} b - c_1, b - c_2 & -c_1, b \\ b, -c_2 & 0, 0 \end{pmatrix} \qquad M_2 = \begin{pmatrix} b - c_1, b - c_2 & -c_1, 0 \\ 0, -c_2 & 0, 0 \end{pmatrix} \tag{2}$$

For M_1: $\delta_1 = c_1$ and $\delta_2 = c_2$ however for M_2, $\Delta f_1((1,1)) = b - c_1$ but $\Delta f_1((1,0)) = -c_1$. The Prisoner's Dilemma (as defined by M_1) is *state-independent* but the Stag-hunt game (as defined by M_2) is not. Further games considered in [1] are the Volunteer's dilemma and the Volunteer's timing dilemma (a generalisation with more than 2 strategies) which are both not *state-independent.*

To define introspection dynamics we introduce the following two quantities:

- **Fermi function.** Each player i has a personal Fermi function $\phi_i(x) = (1 + e^{\beta_i x})^{-1}$. Since $\beta_i > 0$, the exponential $e^{\beta_i x}$ is strictly increasing in x, so ϕ_i is strictly decreasing with $\phi_i(0) = \frac{1}{2}$ and $\phi_i(x) + \phi_i(-x) = 1$.
- **Mutation.** Each player i has mutation probabilities $\mu_{i0}, \mu_{i1} > 0$ with $\mu_{i0} + \mu_{i1} < 1$, where μ_{i0} is the probability of mutating to cooperation and μ_{i1} is the probability of mutating to defection, regardless of the intended choice of the dynamic.

Introspection dynamics. At each step, one player i is selected uniformly at random. Player i considers switching from action a_i to the alternative $\bar{a}_i = 1 - a_i$. The transition probability for the switch is

$$T^{\mathrm{intro}}_{\mathbf{a},\,\mathbf{a}_{i \to \bar{a}_i}} = \frac{1}{N}\big[(1 - \mu_{i0} - \mu_{i1})\phi_i(\Delta f_i(\mathbf{a})) + \mu_{i,\bar{a}_i}\big], \tag{3}$$

where $\mu_{i,\bar{a}_i}$ denotes μ_{i0} when $\bar{a}_i = C$ and μ_{i1} when $\bar{a}_i = D$. Since ϕ_i is strictly decreasing, a larger payoff gain from keeping the current action yields a smaller switching probability.

Definition 1 (Cooperation probability)**.** *Given an ergodic chain on S with stationary distribution π, the* cooperation probability *is $p_C = \pi \cdot s$, where $s_{\mathbf{a}} = \frac{1}{N}\sum_i a_i$ is the fraction of cooperators in state $\mathbf{a}$.*

3 Individual cooperation probabilities

For a state-independent game, $\Delta f_i(\mathbf{a})$ depends only on a_i, not on $\mathbf{a}_{-i}$. We use this to compute each player's long-run cooperation probability and the stationary distribution directly from the two-state update rule.

Theorem 2 (Individual cooperation probabilities)**.** *For any state-independent game with constants δ_i, under introspection dynamics with player-specific choice intensity β_i and mutation probabilities $\mu_{i0}, \mu_{i1} > 0$ with $\mu_{i0} + \mu_{i1} < 1$, the long-run probability that player i cooperates is*

$$p_i = \phi_i(\delta_i)(1 - \mu_{i0} - \mu_{i1}) + \mu_{i0}. \tag{4}$$

Proof. By state-independence, $\Delta f_i(\mathbf{a}) = (1 - 2a_i)\delta_i$ depends only on a_i. When player i is selected, the probability their next action is C equals p_i regardless of their current action: if $a_i = D$, the switch probability is $\phi_i(\delta_i)(1 - \mu_{i0} - \mu_{i1}) + \mu_{i0} = p_i$; if $a_i = C$, the probability of remaining at C is $1 - [\phi_i(-\delta_i)(1 - \mu_{i0} - \mu_{i1}) + \mu_{i1}] = p_i$, using $\phi_i(x) + \phi_i(-x) = 1$. Since each selection of player i places them in state C with probability p_i independently of their current state, the long-run cooperation probability of player i is p_i.

□

Theorem 3 (Stationary distribution)**.** *For any state-independent game with constants δ_i, the stationary distribution of the introspection chain on $S = \{0,1\}^N$ is the product measure*

$$\pi_{\mathbf{a}} = \prod_{i=1}^{N} p_i^{a_i}(1 - p_i)^{1 - a_i}, \qquad \mathbf{a} \in S, \tag{5}$$

where p_i is given by (4)*.*

Proof. By Theorem 2, each selection of player i places them in state C with probability p_i independently of every other player's action. In stationarity the players' actions are therefore independent, with player i cooperating with probability p_i and defecting with probability $1 - p_i$. The probability of state $\mathbf{a}$ is the product of these marginals, giving (5).

□

4 State-independence of the heterogeneous public goods game

The *heterogeneous public goods game* is played by N players with player-specific contributions $\alpha_1, \ldots, \alpha_N > 0$ and public goods multipliers $r_1, \ldots, r_N > 1$. The payoff to player i in state $\mathbf{a} = (a_1, \ldots, a_N)$ is

$$f_i(\mathbf{a}) = \frac{r_i}{N} \sum_{j=1}^{N} \alpha_j a_j - \alpha_i a_i. \tag{6}$$

We write $C(\mathbf{a}) = \sum_j \alpha_j a_j$ for the total contribution.

Lemma 4 (State-independence of the PGG)**.** *The payoff* (6) *is state-independent for every player i, with*

$$\delta_i = \alpha_i \left(1 - \frac{r_i}{N}\right), \tag{7}$$

i.e. $\Delta f_i(\mathbf{a}) = (1 - 2a_i)\, \alpha_i (1 - r_i/N)$, *independent of* $\mathbf{a}_{-i}$.

Proof. When player i switches from a_i to $\bar{a}_i = 1 - a_i$, only their own contribution changes, so the total contribution becomes $C(\mathbf{a}_{i \to \bar{a}_i}) = C(\mathbf{a}) + \alpha_i(1 - 2a_i)$. Expanding both payoffs:

$$\begin{aligned} f_i(\mathbf{a}) &= \frac{r_i}{N}\, C(\mathbf{a}) - \alpha_i a_i, \\ f_i(\mathbf{a}_{i \to \bar{a}_i}) &= \frac{r_i}{N} \big[C(\mathbf{a}) + \alpha_i(1 - 2a_i)\big] - \alpha_i(1 - a_i). \end{aligned}$$

Subtracting:

$$\begin{aligned} \Delta f_i &= -\frac{r_i}{N} \alpha_i (1 - 2a_i) + \alpha_i(1 - 2a_i) \\ &= \alpha_i(1 - 2a_i)\left(1 - \frac{r_i}{N}\right). \end{aligned}$$

The term $C(\mathbf{a})$ cancels entirely, confirming state-independence with $\delta_i = \alpha_i(1 - r_i/N)$. □

Corollary 5 (Exact cooperation probability)**.** *For the heterogeneous public goods game* (6)*, the long-run cooperation probability is*

$$\boxed{p_C^{\text{intro}} = \frac{1}{N} \sum_{i=1}^{N} \left[\frac{1 - \mu_{i0} - \mu_{i1}}{1 + e^{\beta_i \alpha_i (1 - r_i/N)}} + \mu_{i0} \right].} \tag{8}$$

Proof. By Lemma 4, $\delta_i = \alpha_i(1 - r_i/N)$. By Theorem 2: $p_C^{\text{intro}} = \frac{1}{N} \sum_{i=1}^{N} p_i$. Substituting $\phi_i(\delta_i) = (1 + e^{\beta_i \alpha_i (1 - r_i/N)})^{-1}$ into (4) gives (8).

□

Figure 1 validates Corollary 5 against exact values and simulation across group sizes. Table 1 verifies Theorem 3: for $N = 3$ with per-player multipliers $r_1 = 1$, $r_2 = 3 = N$, $r_3 = 9$, uniform contributions $\alpha_i = 1$, $\beta = 2$, and $\mu_{i0} = \mu_{i1} = 0.1$, the individual cooperation probabilities are $p_1 \approx 0.267$, $p_2 = 0.500$, $p_3 \approx 0.886$, and the formula and exact values of $\pi_{\mathbf{a}}$ agree to five decimal places for all eight states.

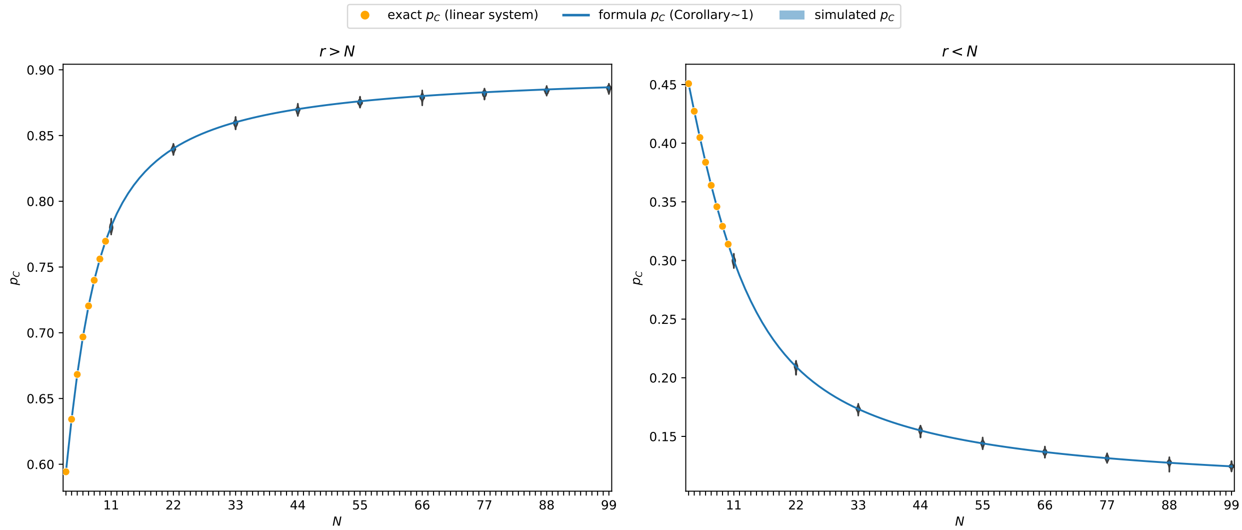

Figure 1: Validation of formula (8) on a heterogeneous group with $\alpha_i = i$, $\beta = 0.5$, $\mu_{i0} = \mu_{i1} = 0.1$. *Left:* $r = 2N$ (so $r > N$ for all N). *Right:* $r = N/2$ (so $r < N$ for all N). Orange dots: exact p_C from the full 2^N linear system (feasible only for small N). Blue curve: formula (8) (Corollary 5). Blue bands: p_C estimated from 10^5 simulation steps across 99 independent runs. All values obtained using [3].

5 Structural consequences

The formulas (4) and (8) allow immediate structural conclusions about individual and aggregate cooperation.

Corollary 6 (Player-specific cooperation thresholds)**.** *Player i cooperates with probability greater than $\frac{1}{2}$ if and only if*

$$\phi_i(\delta_i) > \frac{\frac{1}{2} - \mu_{i0}}{1 - \mu_{i0} - \mu_{i1}}.$$

For the PGG, $\delta_i = \alpha_i(1 - r_i/N)$; when $\mu_{i0} = \mu_{i1}$ this reduces to $r_i > N$. A sufficient condition for $p_C^{\text{intro}} > \frac{1}{2}$ is that the inequality holds for every i.

Proof. Since $1 - \mu_{i0} - \mu_{i1} > 0$:

$$p_i > \tfrac{1}{2} \iff (1 - \mu_{i0} - \mu_{i1})\phi_i(\delta_i) > \tfrac{1}{2} - \mu_{i0} \iff \phi_i(\delta_i) > \frac{\frac{1}{2} - \mu_{i0}}{1 - \mu_{i0} - \mu_{i1}}.$$

When $\mu_{i0} = \mu_{i1}$, the right-hand side equals $\frac{1}{2}$, and since ϕ_i is strictly decreasing with $\phi_i(0) = \frac{1}{2}$, this is equivalent to $\delta_i < 0$. For the PGG, $\delta_i = \alpha_i(1 - r_i/N) < 0$ iff $r_i > N$. The claim about p_C^{intro} follows by averaging.

□

Corollary 7 (Neutral drift)**.** *If $\beta_i = 0$ for all i, then ϕ_i is constant and the payoff parameters δ_i have no effect on p_C; cooperation is determined entirely by mutation:*

$$p_i = \frac{1 + \mu_{i0} - \mu_{i1}}{2}, \qquad p_C^{\text{intro}} = \frac{1}{2} + \frac{1}{2N}\sum_{i=1}^{N}(\mu_{i0} - \mu_{i1}).$$

State $\mathbf{a}$	Formula $\pi_{\mathbf{a}}$	Exact $\pi_{\mathbf{a}}$
DDD	0.04193	0.04193
DDC	0.32463	0.32462
DCD	0.04193	0.04193
DCC	0.32463	0.32462
CDD	0.01526	0.01527
CDC	0.11818	0.11818
CCD	0.01526	0.01527
CCC	0.11818	0.11818

Table 1: Stationary-distribution probabilities $\pi_{\mathbf{a}}$ for all eight states $\mathbf{a} \in \{D, C\}^3$ with $N = 3$, $\alpha_i = 1$, $\beta = 2$, $\mu_{i0} = \mu_{i1} = 0.1$, and per-player multipliers $r_1 = 1 < N = r_2 = 3 < r_3 = 9$. Formula: product measure (5) (Theorem 3); Exact: values from the 2^N linear system obtained using [3]. Both columns to five decimal places.

In particular, $p_C^{\text{intro}} = \frac{1}{2}$ *if and only if* $\sum_i (\mu_{i0} - \mu_{i1}) = 0$.

Proof. When $\beta_i = 0$, the Fermi function satisfies $\phi_i(x) = \frac{1}{2}$ for all x. Substituting into (4):

$$p_i = \frac{1 - \mu_{i0} - \mu_{i1}}{2} + \mu_{i0} = \frac{1}{2} + \frac{\mu_{i0} - \mu_{i1}}{2} = \frac{1 + \mu_{i0} - \mu_{i1}}{2}.$$

□

Corollary 8 (Role of heterogeneity)**.** *For the PGG, for player i with $r_i \neq N$:*

(i) $\dfrac{\partial p_i}{\partial \alpha_i}$ *is positive when $r_i > N$ and negative when $r_i < N$.*

(ii) $\dfrac{\partial p_i}{\partial \beta_i}$ *is positive when $r_i > N$ and negative when $r_i < N$.*

Proof. Write $p_i = (1 - \mu_{i0} - \mu_{i1})\phi_i(\alpha_i(1 - r_i/N)) + \mu_{i0}$.

Part (i). Differentiating with respect to α_i:

$$\frac{\partial p_i}{\partial \alpha_i} = (1 - \mu_{i0} - \mu_{i1})(1 - r_i/N)\, \phi_i'(\alpha_i(1 - r_i/N)).$$

Since $\phi_i'(x) < 0$ for all x and $1 - \mu_{i0} - \mu_{i1} > 0$, the sign equals $-\mathrm{sgn}(1 - r_i/N) = \mathrm{sgn}(r_i/N - 1)$, which is positive if and only if $r_i > N$.

Part (ii). Write $g(t) = (1 + e^t)^{-1}$, so $\phi_i(x) = g(\beta_i x)$. Differentiating with respect to β_i:

$$\frac{\partial p_i}{\partial \beta_i} = (1 - \mu_{i0} - \mu_{i1})\, \alpha_i(1 - r_i/N)\, g'(\beta_i \alpha_i(1 - r_i/N)).$$

Since $g' < 0$ and $\alpha_i > 0$, the sign equals $-\mathrm{sgn}(1 - r_i/N) = \mathrm{sgn}(r_i/N - 1)$, which is positive if and only if $r_i > N$. □

Remark 9. *Corollaries 6–8 show that each player's long-run cooperation is fully determined by their individual triple (α_i, β_i, r_i) and the group size N. Players with $r_i > N$ cooperate more than half the time, and increasing α_i or β_i raises their cooperation probability further; players with $r_i < N$ show the opposite sensitivity. The parameters of other players have no effect on player i's marginal cooperation probability.*

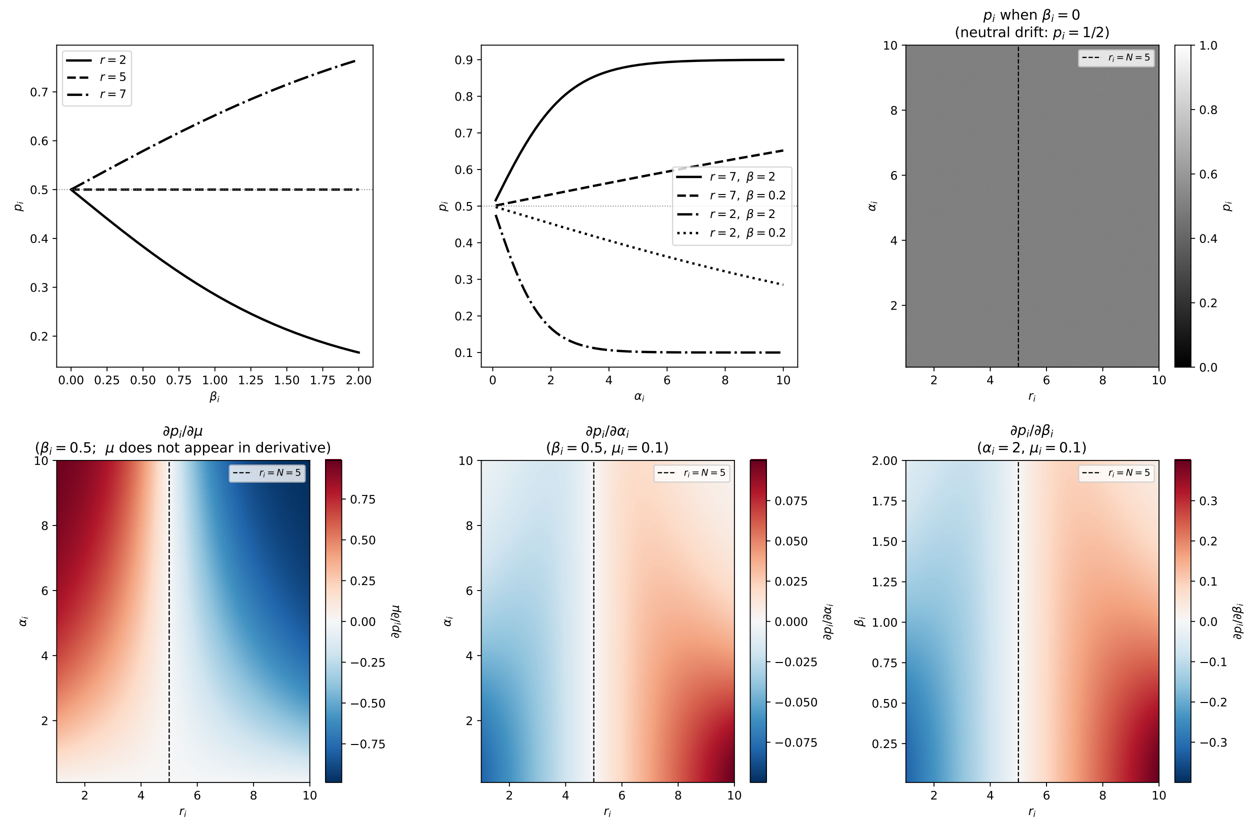

Figure 2: Structural properties of p_i for a single player with $N = 5$, $\mu_{i0} = \mu_{i1} = 0.1$. *Top left:* p_i as a function of β_i ($\alpha_i = 2$) for $r < N$, $r = N$, and $r > N$; all curves pass through $p_i = \frac{1}{2}$ at $\beta_i = 0$ (Corollary 7). *Top centre:* p_i as a function of α_i for two values of r and β_i; curves above (below) $\frac{1}{2}$ correspond to $r > N$ ($r < N$). *Top right:* $p_i(\alpha_i, r_i)$ at $\beta_i = 0$, confirming $p_i = \frac{1}{2}$ everywhere (Corollary 7); the dashed line marks $r_i = N$. *Bottom row:* Heatmaps of the partial derivatives $\partial p_i/\partial\mu_i$, $\partial p_i/\partial\alpha_i$, and $\partial p_i/\partial\beta_i$ over the (r_i, α_i) or (r_i, β_i) plane ($\beta_i = 0.5$ or $\alpha_i = 2$ as fixed parameter). Red indicates a positive derivative; blue indicates negative. Each derivative changes sign exactly at $r_i = N$ (dashed line), consistent with Corollary 8.

6 Conclusion

We have proved that introspection dynamics on any state-independent game is exactly solvable. When Δf_i depends only on a_i, each selection of player i places them in state C with probability p_i regardless of current state (Theorem 2), and the stationary distribution is a product measure (Theorem 3). For the heterogeneous public goods game, the linear payoff structure implies state-independence (Lemma 4), and the long-run cooperation probability has a closed form with N terms rather than 2^N (Corollary 5).

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